

Section A (80 marks)

Answer all questions in the space provided

Question 1

Fig. 1.1 shows a cell undergoing two different stages of meiosis (to the same scale).

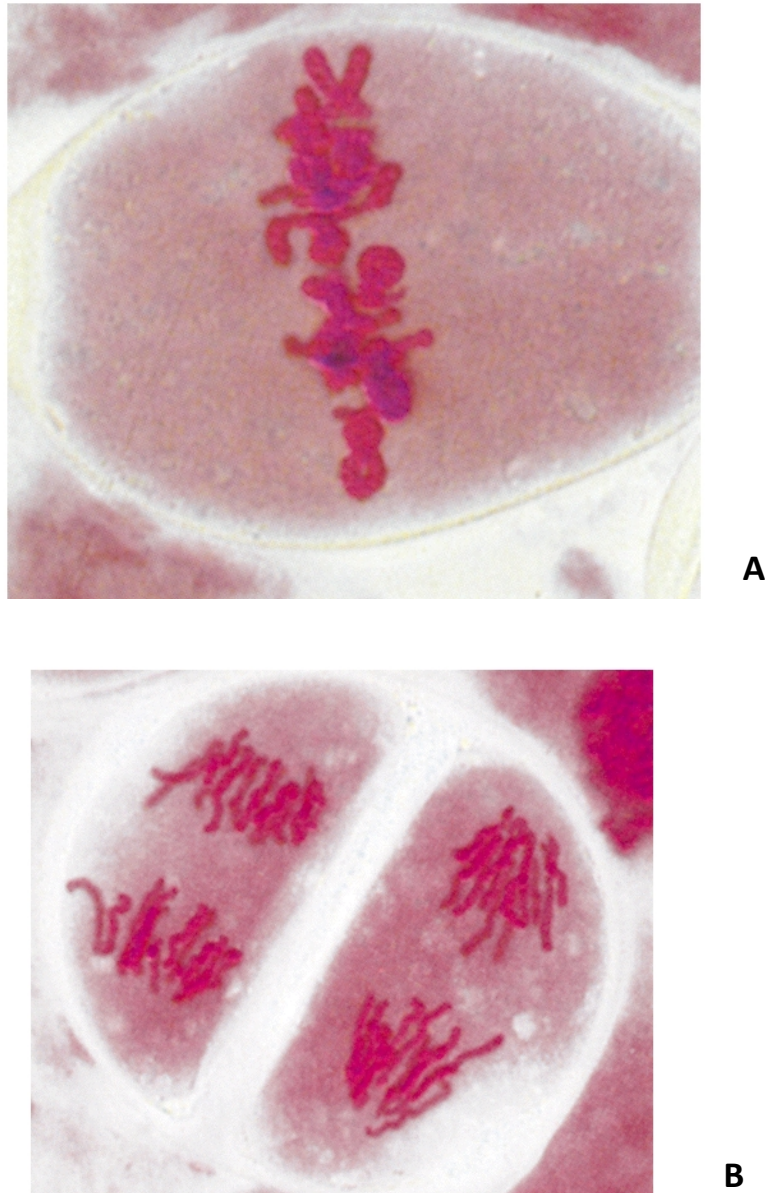


Fig. 1.1

(a) Identify stages **A** and **B**. [1]

- A is **metaphase I**
- B is **anaphase II**.

(b) Explain the phenomenon observed in B. [2]

- **Centromeres divide**
- and the **chromatids separate** and are pulled, centromeres first, by the spindle fibres **to opposite poles**.
- This is due to the **shortening of** the centromere-to-pole **spindle fibres**.

[1 mark for any of the 2 points above]

(c) Describe how meiosis can lead to genetic variation. [2]

- In **Prophase I**, non-sister chromatids of **homologous chromosomes** align themselves alongside one another, and **form chiasma**. **Crossing over** occurs and new combinations of alleles on the chromosomes of the gametes are formed.
- During **Metaphase I** and **Anaphase I** respectively, the **arrangement and subsequent separation of the chromosomes** of each bivalent is completely **independent of** the orientation of the **other bivalents**. This gives rise to different combinations of chromosomes in a gamete.
- During **Metaphase II** and **Anaphase II** respectively, the **arrangement and subsequent separation of the chromatids** of each chromosome is **completely independent of** the orientation of the **other chromosomes**. This gives rise to different combinations of chromosomes in a gamete.

[1 mark for any of the 2 points above]

Okadaic acid is a toxin that causes spindle fibres to significantly increase in length and disrupts the interactions between microtubules and kinetochores, resulting in a disorganized metaphase plate.

(d) Suggest a possible outcome of a disorganised metaphase plate. [1]

- **The chromosomes and chromatids may not separate equally/properly during Anaphase I or II respectively. [0.5]**
- **Meiosis cannot continue and no germ cells are formed / This lead to non-disjunction. [0.5]**

OR

- **The sister chromatids may not separate equally/properly during Anaphase. [0.5]**
- **Mitosis / cell division cannot continue and no daughter cells are form. [0.5]**

The chromosome carrying alleles **F/f** also carries alleles **E/e**, as shown in **Fig 1.2**.

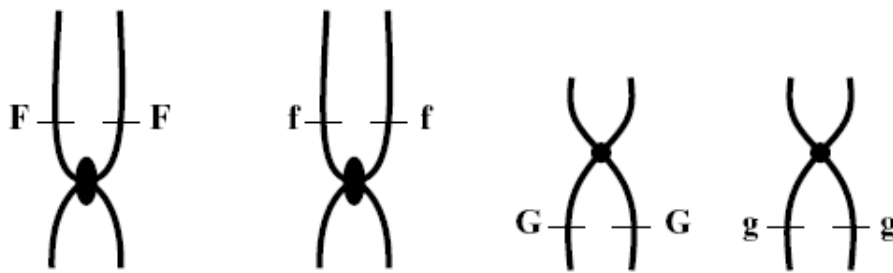


Fig. 1.2

(e)(i) State the genotype of this cell. [1]

FfGg

(ii) Explain how there are two copies of each allele of each gene at this stage of nuclear division. [2]

- **DNA replication has occurred;**
- **During interphase / S-phase of interphase before mitosis;**
- **Giving rise to two chromatids;**
- **With two copies of the allele, one on each chromatid;**

[Total: 9 marks]

Question 2

Fig. 2.1 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it.

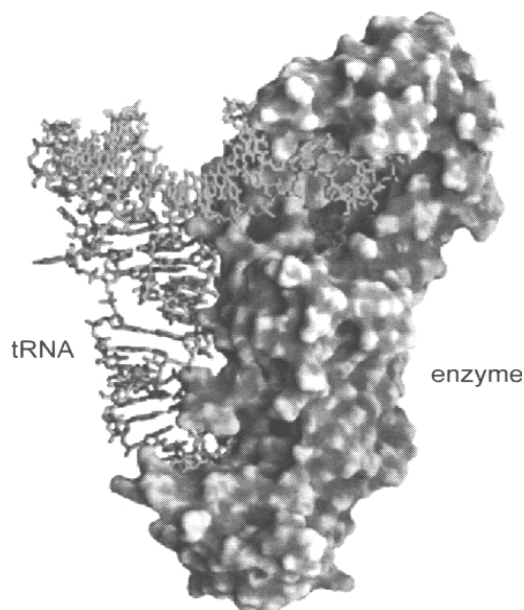


Fig. 2.1

(a) Describe **two** structural features which adapt tRNA to its role in translation. [2]

1 3' / (CCA end) = allows specific amino acid molecule binds to the tRNA

2 loop with anticodon /region has specific triplet base sequence - complementary to a codon on mRNA.

(b) Explain how the enzyme in **Fig. 2.1** is suited to its function. [3]

1. Identify the enzyme as aminoacyl-tRNA synthetase
2. has recognition/active site for both amino acid
3. and tRNA with specific anticodon;
4. with site being complementary in shape and charge to the substrates
5. able to catalyse formation of covalent bond between tRNA and amino acid
6. twenty different aminoacyl-tRNAase enzymes for each amino acid to ensure specificity in binding

(c) **Fig. 2.2** below shows the base sequence on a length of mRNA. The key indicates which amino acids they represent.

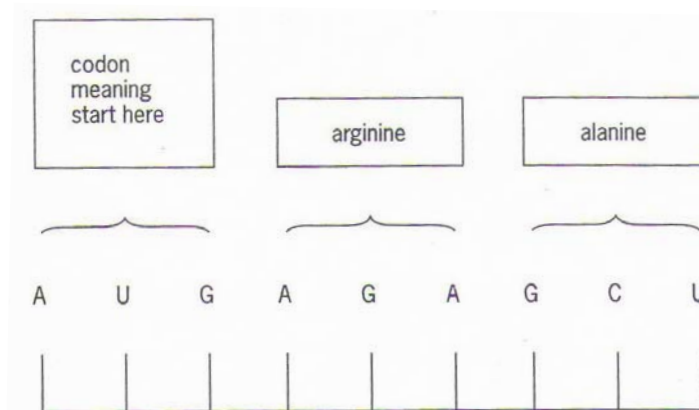


Fig. 2.2

The amino acid arginine is shown here as AGA but can also be represented by the codon AGG.

(i) Explain why the substitution of A by G in this case does not affect the amino acid represented. [1]

Genetic code is degenerate/redundancy in genetic code;

1 amino acid can be coded by more than 1 codon;

First two bases are most important in determining which amino acid is attached;

The third base is a wobble base/not as important;

It is a form of silent mutation

1 max

- (ii) The codons in the mRNA sequence do not overlap. Suggest 2 advantages of this. [2]

Substitution of a base will only change one amino acid instead of two;;
Overlapping can cause difficulties in translation when the tRNA comes into the P site as it would be blocked by the previous tRNA;;
There is less restriction on the order of amino acids since the overlapped base of one codon has no effect on the next codon;;

Max 2

Some genetic drugs are short sequences of nucleotides. They act by binding to selected sites on DNA or mRNA molecules and preventing the synthesis of disease-related proteins. There are two types:

1. triplex drugs are made from DNA nucleotides and bind to the DNA forming a three stranded helix.
2. antisense drugs are made from RNA nucleotides and bind to mRNA.

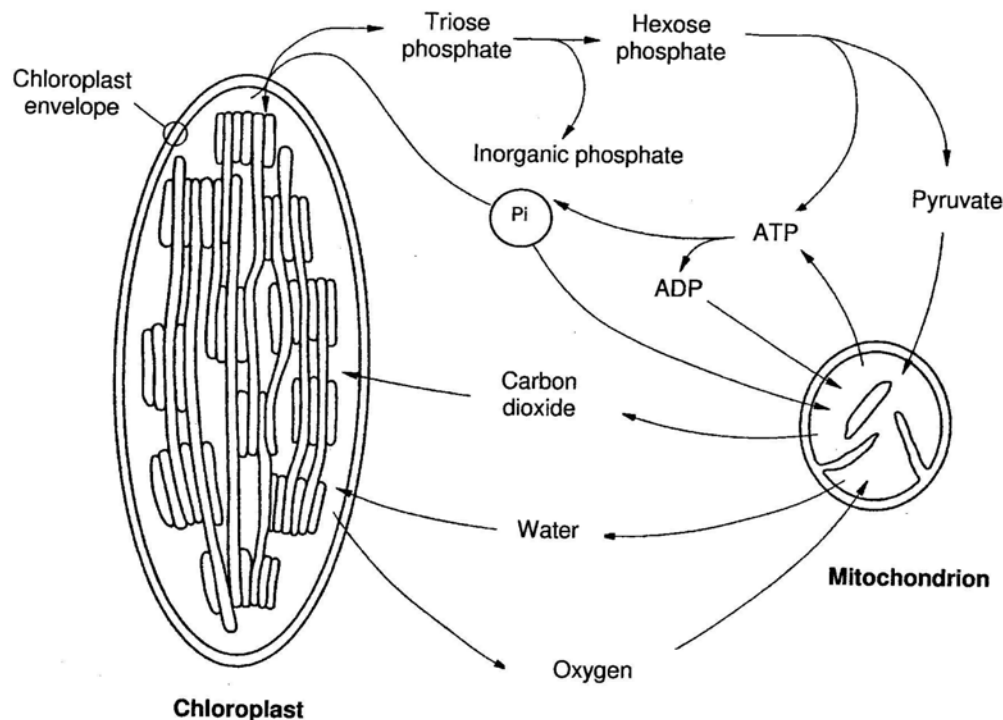
- (iii) Name the process in protein synthesis which will be inhibited by [1]:

Drug type	Process inhibited
Triplex drugs	Transcription
Antisense drugs	Translation

[Total: 9 marks]

Question 3

Fig. 3.1 shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.



A leafy shoot is sealed inside an air-tight container. The concentration of oxygen in the atmosphere within this container is measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Fig. 3.1**, outline how the following links the metabolic pathways between the chloroplast and mitochondrion of this cell.

- (i) triose phosphate and hexose phosphate; [2]

in stroma of chloroplast, triose phosphate (TP) is formed as a product of light independent / dark reaction;(1m) TP moves out of chloroplast into the cytosol and is converted to hexose phosphate (1m)

that can enter glycolysis as a respiratory substrate, to be eventually oxidized during aerobic respiration into pyruvate (1m) which enters the mitochondria to undergo the link reaction / to be converted to acetyl CoA(1m) max [2]

- (ii) gaseous exchange. [2]

In granum of chloroplast, oxygen is formed as a product of photolysis of water during the light dependent/ light reactions (1m); oxygen diffuse out of chloroplast and act as the final electron acceptor in the electron transport chain (ETC) in mitochondrial matrix(1m); CO₂ released during Krebs cycle/ link reaction released into cytosol which diffuse into stroma of chloroplast and used in dark / light independent reaction/ carbon fixation (1m)
max [2]

- (b) With reference to **Fig. 3.1**,

- (i) explain how it is possible for the oxygen concentration to remain constant. [1]

Oxygen released by chloroplast during the light dependent reaction of photophosphorylation/ photolysis of water is used by mitochondrion as final electron acceptor, compensation point has been reached(1m) Rate of respiration equal to rate of photosynthesis(1m), thus no net change in oxygen concentration.(1m)
max [1]

- (ii) chloroplasts do not have the transporters that allow them to export ATP to the cytosol. How then does the rest of the cell get the ATP it needs for cellular metabolism? [2]

ATP derived from mitochondria / oxidative phosphorylation are supplied to the cell to meet its need(1m) ; triose phosphate from light-independent/ dark reaction will move to cytosol and may eventually be used as respiratory substrate(1m) to drive glycolysis and thus generate ATP.(1m)
max [2]

In plants, the carbohydrates manufactured are channeled to form two types of macromolecules, namely cellulose and starch.

- (c) State the two structural differences between cellulose and starch. [2]

- Cellulose basic units β glucose while starch is composed of α glucose
- Cellulose is composed of straight chains polysaccharide while starch made of polysaccharide chains that are coiled into helices
- Cellulose has hydrogen bonding between adjacent chains while no such hydrogen bonds exist between the polysaccharide chains.
- Cellulose polysaccharide chains are unbranched while in starch (amylopectin), there is branching (glycosidic bond is between carbons 1 and 6)

1 mark per point/ max [2]

- (d) Explain how these differences allow them to serve different functions in a plant cell. [2]

Cellulose: long straight chains allow for stacking and formation of hydrogen bonds to form bundles of fibrils to give tensile strength (1m) which is necessary in plant cells to give support and structure (1m).

Starch is coiled and compact to serve as storage material (1m), occupying less space and is insoluble so it does not affect osmotic potential of cell, (1m)

1 mark per point/ max [2]

[Total: 11 marks]

Question 4

In an experiment, pea plants with round, black seeds were crossed with pure-breeding pea plants with wrinkled, green seeds. The offspring all had round, black seeds. These seeds were grown and the resultant plants allowed to self-pollinate (fertilise).

- (a)(i) What would be the expected phenotypic ratio? Using appropriate symbols draw a genetic diagram to explain your results. [5]

The above cross in (a)(i) produced 324 offspring with the following characteristics, as shown in **Table 4.1**.

Number of offspring	Phenotype
181	round, black seeds
51	round, green seeds
53	wrinkled, black seeds
39	wrinkled, green seeds

Table 4.1

- (ii) Complete the table below, **Table 4.2**, and calculate the chi-square to test the significance of the differences between the observed and expected results. [2]

	round, black seeds	round, green seeds	wrinkled, black seeds	wrinkled, green seeds
Observed number (O)	181	51	53	39
Expected number (E)	182.25	60.75	60.75	20.25
Difference between observed & expected numbers (O – E)	-1.25	-9.75	-7.75	18.75
$(O - E)^2$	0.00517.36	95.0625	60.0625	351.5625
$(O - E)^2 / E$	0.0008573	1.5648	0.9886	17.361
$\chi^2 = \Sigma [(O - E)^2 / E]$	19.922 round off 19.9			

Table 4.2

	Probability, p				
Degrees of freedom	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Table 4.3

- (iii) With reference to the table of probabilities, **Table 4.3**, explain how the value for the chi-squared test supports the hypothesis that these are two pairs of segregating alleles at two loci. [2]

-Since value of chi-square calculated 19.9 > critical value 7.82, the null hypothesis is rejected.

-Result of the cross does not support null hypothesis is that the shape and colour of peas are two pairs of segregating alleles at 2 loc /the 2 genes are not linked

Question 5

With the completion of the Human Genome Project, it was estimated that the human genome was likely to contain only 30,000 genes. Counting the number of protein-encoding genes in the human genome is a difficult process, as the “start” and “stop” boundaries of transcribed genes are only imperfectly understood.

A young scientist has decided to take a different approach of capturing mRNAs from the cytoplasm, before converting them to DNA to estimate the number of genes present. Using this approach, he reports that there are over 65,000 discrete mRNA-transcribing sequences in the genome, hence there are more genes than estimated by the Human Genome Project.

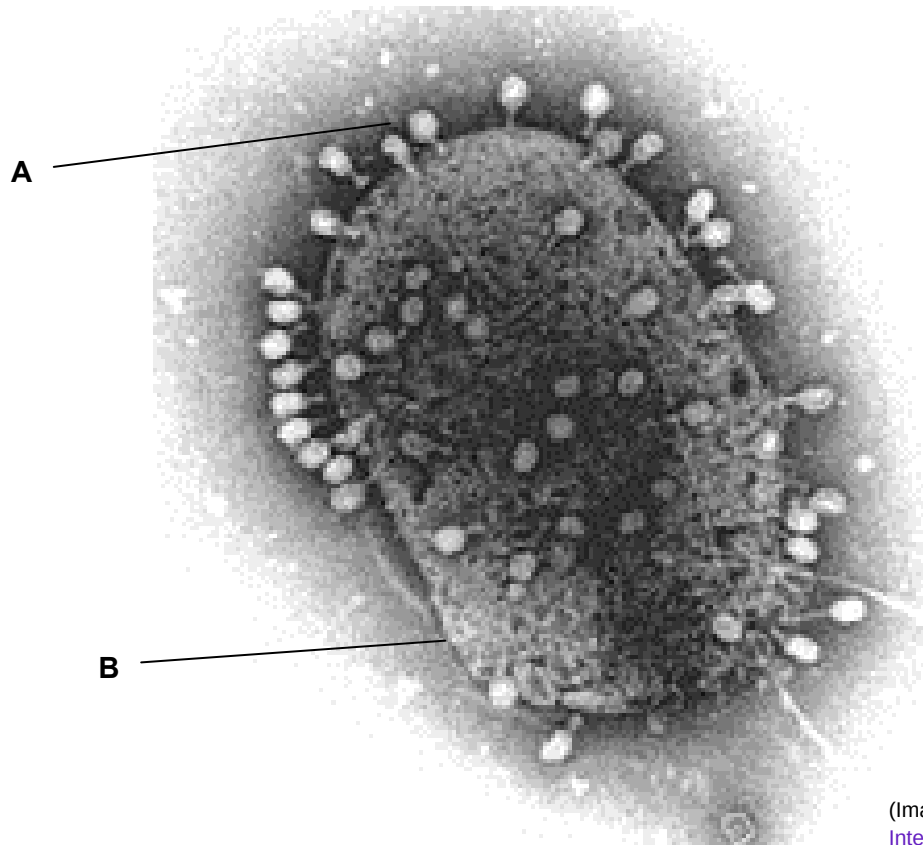
- (a) Describe briefly how post-transcriptional modification would allow all types of mRNA to be captured using a single method, and how mRNA can be subsequently converted to DNA. [2]
- **addition of a poly-A tail to pre-mRNA / polyadenylation**
 - **allows allow mRNA to be captured by use of the single-stranded poly-T sequence bound to a column matrix**
 - **after the bound mRNA is eluted from the matrix, the mRNA can be added to reverse transcriptase**
 - **to catalyse the formation of cDNA**
- (b) State why counting the number of unique mRNA sequences will result in an over-estimate of the number of actual genes in an organism. [1]
- **One gene may code for more than one type of mRNA**
 - **Due to alternative splicing**
 - **Where different exons are spliced together**
 - **To give rise to different mRNA sequences**

- (c) Within the same human cell, the level of gene expression can differ greatly between genes. Briefly describe what could account for the different levels of mRNA produced for different genes. [2]
- Different levels of condensation of DNA / ref. genes within the same cell could be located in either euchromatin or heterochromatin regions
 - Due to different extents of histone modification and DNA methylation
 - Hence different rate of transcription of the genes by RNA polymerase to form different levels of mRNA.
 - Also due to presence of activators and repressors in the cell
 - Binding to enhancers and silencers respectively
 - to upregulate or downregulate the expression of a gene.
- (d) The eukaryote genome differs from the prokaryote genome in that some genes may be present in multiple copies, a result of gene amplification.
- (i) Describe briefly one process by which gene amplification can occur. [2]
- Unequal crossing over of segments of non-sister chromatids of homologous chromosomes leading to gene duplication
 - Presence of identical repeated sequences presents a problem for precise pairing in subsequent meiosis which could lead to further unequal crossing-over events to give a string of the same gene in a row.
 - Or Slippage occurring during DNA replication such that the template shifts with respect to the new complementary strand
 - And one region of the template is copied twice.
 - Or Occur via the production of an mRNA transcript of the to-be-amplified gene sequence, and
 - By the reverse transcription of the mRNA to produce an additional copy of the original DNA sequence.
- (ii) In nature, some organisms are able to amplify certain genes at some stage in their development. Describe the significance of gene amplification in development. [2]
- Gene amplification in the cells of an organism increases the number of templates for transcriptions;
 - results in increased gene expression/ Allows more copies of mRNA to be made
 - increase amt of gene product in a short period of time;
 - more polypeptide synthesised to carry out specialised functions in the cell/ for construction of certain structures of the cell;
 - Eg. Amplification of the rRNA genes early in cell development of the frog
 - Ensures that sufficient ribosomes can be made
 - For increased rate of protein synthesis to synthesise the proteins needed in large amts of that particular developmental stage

[Total: 9 marks]

Question 6

Fig. 6.1 shows a scanning electron micrograph of *Escherichia coli* under mass attack by numerous phage-T4 virions.



(Image courtesy of [Cornell Integrated Microscopy Center](#))

Fig. 6.1

- (a)(i) Suggest how **A** recognizes the correct host cell. [1]
- (ii) With reference to **Fig. 6.1**, describe a process whereby bacterial DNA can be transferred from one cell to another. [2]

- By generalized transduction;;
- A/ T4 phages undergoes lytic cycle and degrades/ breaks down B's/ bacterial DNA;
- When new T4 phages are being assembled, occasionally, fragments of host's DNA or even entire plasmid;
- can be packaged randomly into viral capsid to form new phage particles;
- When resultant T4 phages infect another bacterium as shown by Figure 4.1;
- They inject the donor DNA/ viral DNA and bacterial DNA from first bacterium into this bacterium enabling the bacterial DNA from first bacterium to be transferred to the second cell.

- This does not initiate a lytic cycle but genetic recombination occur for second bacterium/ some of the donor DNA from the first bacterium may replace the homologous regions of the second bacterium's chromosome/ reference to crossing over;

(iii) Give **two** differences between **2 other processes** (besides that one named in (aii)) that result in genetic variation in bacteria. [3]

Feature	Named Process 1: Conjugation	Named Process 2: Transformation
1. Source of DNA	DNA transferred is F plasmid from a donor bacteria cell/ F ⁺ cell.	DNA transferred is naked DNA from the environment/ either from dead bacteria that released DNA fragments or artificially introduced into surrounding.
2. Requirement	It involves a donor and recipient cells in direct contact with a cytoplasmic bridge/ sex pilus formed between them.	It involves any competent bacterial cell that has the ability to incorporate naked DNA/ uptake DNA from the surrounding.
3. Nature of genetic material transferred	Genetic material transferred is from closely related bacteria.	Genetic material transferred may be from any other bacterial species / may be foreign and genetically manipulated.

Fig. 6.2 shows the Jacob and Monod model of gene expression in *Escherichia coli*, the lac operon model.

(Source:

<http://www.ccrnp.ncifcrf.gov/~toms/genetic.control/www.uic.edu/lect15.htm>)

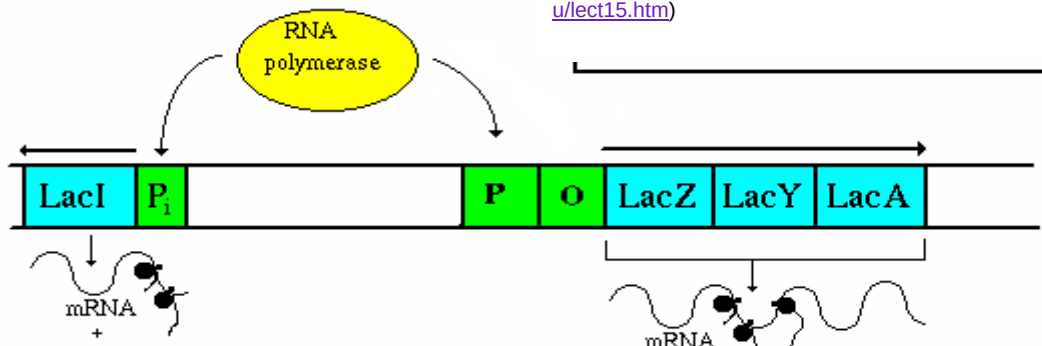


Fig. 6.2

(b) Explain the following terms:

(i) RNA polymerase. [1]

- Enzyme that catalyses
- the synthesis of mRNA/ formation of phosphodiester bonds between two adjacent ribonucleotides/ transcription from DNA template

(ii) Operator O. [1]

- nucleotide/ DNA sequence that binds a repressor molecule
- to control the access of RNA polymerase to the structural genes

(c) With reference to **Fig. 6.2**, distinguish clearly between a regulatory gene and a structural gene. [2]

- **LacI** is a regulatory gene
- which codes for the protein I, a repressor molecule that binds to the operator O
- **LacZ** is a structural gene
- which codes for the enzyme Z, i.e. β -galactosidase that breaks down lactose into glucose and galactose.

(d) Are the enzymes produced by the lac operon inducible or repressible? Explain your answer. [2]

- the lac operon produces inducible enzymes;
- When bacteria is in a lactose medium, the lactose acts as an inducer;
- by combining with the repressor molecule I causing it to be inactive;
- The repressor molecule changes its shape so that it cannot bind to the operator O;
- RNA polymerase can then bind to the promoter;
- and starts the transcription of the structural genes LacZ, LacY and LacA
- enzymes Z, Y and A are hence only produced in response to lactose

[Total: 12 marks]

Question 7

Fig. 7.1 shows two motor neurons, **A** and **B**. It also shows the synapses of neurone **B** with two other neurons, **P** and **Q**.

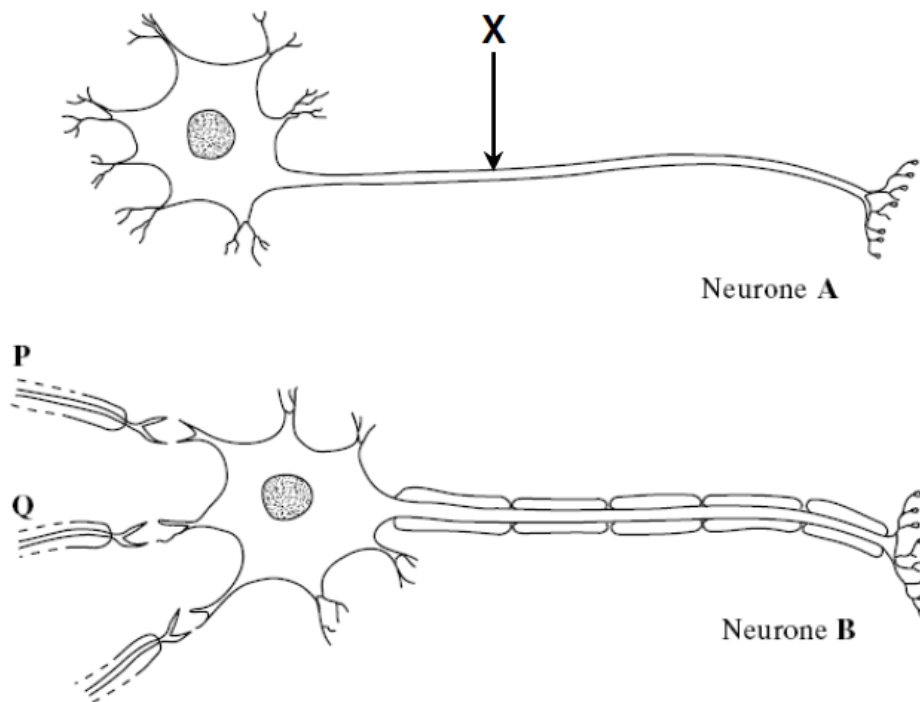


Fig. 7.1

- (a)(i) If an action potential is initiated at point **X** on neuron **A**, describe how this action potential is propagated along the neuron. [3]

local currents are set up on axon memb at adj portion of T, causing _ permeability to Na^+ ;
influx of Na^+ occurs & A.P. is set up once threshold is reached;
local currents are set up btw this region of axon & adj region so that A.P. is propagated away from T;

- (ii) Suggest why transmission of impulses along neuron **B** uses less energy than transmission along neuron **A**. [2]

A.P. jumps from node to node (saltatory conduction) in neurone B (myelinated) & active transport of ions occurs at Node of Ranvier only;
to generate the concentration gradient necessary for facilitated diffusion of Na^+ & K^+ down their respective concentration gradient through channels;
less Na^+/K^+ pumps are required hence less respiration/ATP required to restore ionic balance;

GABA is a neurotransmitter which inhibits the production of action potentials.

Fig. 7.2 and the **graph 7.3** show how the release of GABA from a presynaptic membrane affects the membrane potential of a postsynaptic membrane.

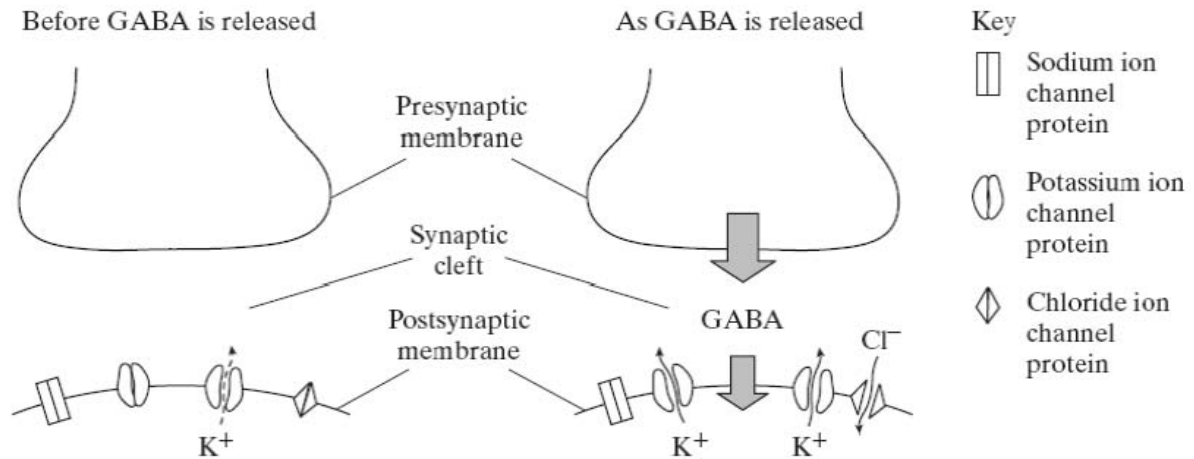
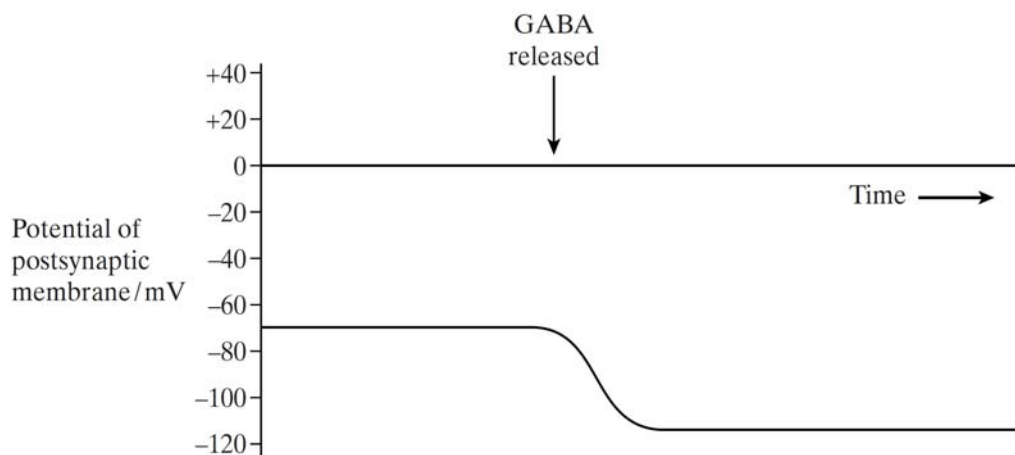


Fig. 7.2



Graph 7.3

- (b) When the postsynaptic membrane is stimulated by acetylcholine, an action potential is less likely to occur if GABA is released at the same time. Explain why. [2]

1. GABA opens K^+ and Cl^- channels; *in psm*
2. More K^+ diffuses out and more Cl^- diffuses into postsynaptic membrane;
3. Membrane potential more negative/hyperpolarized/decreased below resting potential;
4. Requires increased stimulation/must open more Na^+ channels/allow more Na^+ to enter; To reach threshold;

Glucagon, like GABA and acetylcholine, is an essential first messenger molecule in the signal transduction needed in the regulation of blood glucose level in our body. Information from glucagon is transduced as soon as it binds to cellular surface receptors.

- (c)(i) Explain the roles of adenylate cyclase, in signal transduction, that leads to the breakdown of glycogen in liver cells. [2]

**activated adenylate cyclase breakdowns ATP into cAMP & cAMP in turn activates other relay proteins;
activation of enzymes in phosphorylation cascade ;
eventually glycogen phosphorylase activated & glycogen broken down into glucose-1-phosphate;**

- (ii) Suggest the importance of protein kinases and phosphatases in phosphorylation cascades. [2]

**activation of one PK by cAMP leads to activation of subsequent inactivated PK until specific cellular response is activated;
PP remove phosphates from activated PK PK becomes inactive again, thus impt in regulation in PK activity;**

[Total: 11 marks]

Question 8

The Galapagos are a very isolated group of islands. Species of birds called finches live on these islands. Scientists think that the different species have evolved from a population of one species which managed to reach the islands thousands of years ago from South American mainland. Scientists used base sequences in the DNA of the finches to work out the evolutionary relationship between the species. **Fig. 8.1** shows the evolutionary relationship.

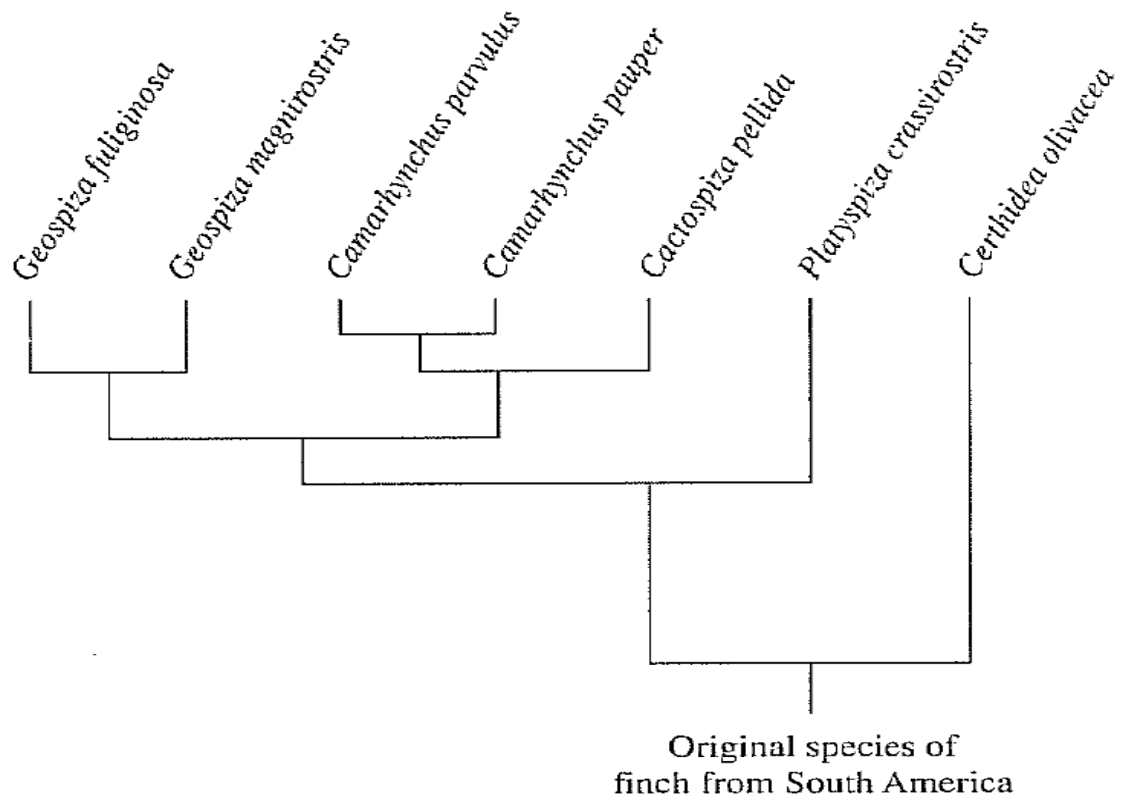


Fig. 8.1. Evolutionary Relationship between the Finches

- (a) Using the information in the diagram, state the present-day species of finch that evolved first on the Galapagos islands? Explain your answer. [2]

Finch species:

- *Certhidea olivacea*;;

Explanation:

- Lineage leading to *Certhidea olivacea* diverged first from the original species of finch from South America;
- Other species diverged later from other ancestries;

(b) Explain why evaluating the DNA base sequences can work out the evolutionary relationship between the species of finches. [2]

- All known life based on nucleic acids;
- Some genes / certain regions of genomes appear to evolve at constant rates;
- Extent of changes to DNA base sequences;
- Determine the extent of relatedness;

(c) The beak of each species is adapted for feeding on a particular type of food. *Geospiza fuliginosa* evolved as an eater of small seeds. *Geospiza magnirostris* evolved on a different island as an eater of large seeds.

(i) Suggest how these different species of finch might have evolved from one species. [3]

- Variation in beak shape present in original population;
- Selection of individuals with beaks suitable for eating small seeds on one island/selection of beak suitable for eating large seeds on another;
- Those with the most suited beaks (survived) to reproduce/pass on their alleles/genes for beak shape;
- Allele frequency changed on each island / populations became genetically different;
- No gene flow between islands/geographical isolation;
- So that individuals from different islands unable to reproduce to produce fertile offspring;

(ii) Describe how biogeography may provide support for the evolution of the different species of finch. [3]

- Biogeography is the geographic distribution of species;
- Species are found where they are because they evolved from ancestors that inhabited those regions;
- The Galapagos are colonized by finch species from the nearest mainland, South America;
- They adapt to their new environments;

[Total: 10 marks]

Section B (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Question 9

(a) Compare and contrast Neo-Darwinism with Neutral Theory. [4]

1. Both Neo-Darwinism (ND) and Neutral Theory (NT) show that Advantages mutations are favored and selected while disadvantages traits are removed during selection.

Differences

Frequency of appearance of genes,

2. **ND in Natural selection** (Selection theory) depends **more on selection** **whereas**
3. in NT frequency of alleles is dependent on **genetic drift**.

Evolution

4. Evolution of population due to **selection of advantageous alleles** as in ND, however
5. in NT evolution is due to mainly **neutral alleles**

Mutations

6. Mutations in ND are defined as **Good/Bad mutations**, however
7. in Neutral theory, mutations are categorized as **Good, Neutral and Bad**

(b) Describe how the membrane affects the entry and exit of substances in and out of a cell.[6]

- **Membrane is selectively permeable, due to the lipid bilayer and membrane proteins**
- **Hydrophobic / non-polar substances are able to dissolve in the lipid bilayer, and pass through the membrane down a concentration gradient**
- **Small molecules such as water and gases are able to pass through the membrane without dissolving**
- **Ions, hydrophilic/polar molecules and large molecules are unable to pass through the fatty acid tails of the bilayer**
- **Presence of trans-membrane proteins which selectively allow certain polar molecules and ions to pass through**
- **May occur down or against a concentration gradient (active transport)**

(c) Define control elements and explain how they influence transcription. [10]

1. **Control elements** are **noncoding DNA segments** that regulate transcription of a gene by binding to proteins like **transcription factors**.
2. Control of gene activity in eukaryotic cells usually involves **a balance between the actions of transcriptional activators and repressors**.
3. Binding of **transcriptional activators** to specific **DNA regulatory sequences called enhancers** turns on transcription.
4. Binding of **transcriptional repressors** to other **DNA regulatory sequences called silencers** turns off transcription.
5. **Enhancers** are a segment of **eukaryotic** DNA containing **multiple control elements**, usually located **far** from the gene whose transcription it regulates.
6. Rate of gene expression is strongly increased by the binding of proteins - **activators** such as **specific transcription factors** to the control elements of enhancers.
7. Activator proteins known as specific transcription factors bind to distal control elements grouped as an enhancer in the DNA.
8. **Protein mediated bending of DNA** bring the **bound activators in contact with mediator proteins** which in turn interact with **general transcription factors at the promoter**.
9. These **multiple protein-protein** interactions help assemble and position the **transcription initiation complex** on the promoter.
10. RNA synthesis is **initiated** and rate of transcription of the gene is **increased**.

Question 10

- (a) Describe the mechanism of synaptic transmission and contrast this with the transmission of a nerve impulse. [6]

Mechanism of synaptic transmission

- Arrival of nervous impulse/AP at the synaptic knob causes depolarization of the presynaptic membrane.
- Depolarization causes calcium channels to open, increases membrane permeability to Ca^{2+} which diffuses into knob down its electrochemical gradient.

Synaptic Vesicles & Acetylcholine

- Ca^{2+} causes synaptic vesicles to fuse with the presynaptic membrane & Synaptic vesicles will empty their contents into the synaptic cleft via exocytosis (ATP required)
- Ca^{2+} are actively pumped out of the synaptic knob after sufficient synaptic vesicles have fused with the presynaptic membrane
- The vesicles will return to the cytoplasm to be refilled with neurotransmitter
- ACh diffuses across the synaptic cleft, delayed by about 0.5ms & attaches to a specific receptor site on the postsynaptic membrane

Excitation

- Arrival of ACh causes a change in the shape of the receptor site which results in the opening of ion channels to allow Na^+ to enter.
- If EPSP reaches threshold, action potential will be initiated in the postsynaptic neurone
- ACh hydrolysed by enzyme acetylcholinesterase, (AChE) aka cholinesterase, to choline and ethanoic acid
- The 2 components will be reabsorbed into synaptic knob (endocytosis) to be recycled into ACh (ATP required)

Synaptic transmission	Transmission along neurone
Chemical in nature	Electrical in nature
Takes place btw two neurones	Takes place along ONE neurone
Synapses can inhibitory or excitatory	Only excitatory

(max 3 mks)

Accept any other differences (there are many differences)

- (b) With reference to the three different stages of cell signaling, describe the series of reactions that take place after a glucagon hormone binds to its target cell, causing the overall increase of blood sugar level. [8]

Signal reception

The hormone, glucagon acts the “first messenger”, binding specifically to a G-protein-linked receptor located on the outside surface of the plasma membrane of the target cell.

Forming a hormone receptor complex

Signal molecule is the ligand

Signal transduction

activates the G protein.

The activated G protein then activates the enzyme adenylyl cyclase,

Adenylyl cyclase catalyses the formation of cAMP from ATP.

cAMP acts as a second messenger

moves within the cell to activate enzymes

such as protein kinases.

Protein kinases then activate other enzymes by phosphorylating them.
In a cascade reaction
Signal is amplified

Cellular response

regulate cellular activities in the cytoplasm
by affecting catalysis by an enzyme or cause the opening or closing of an ion channel
regulate transcription in the nucleus
by synthesis of enzymes or other proteins from genes

(c) Explain the small yield of ATP from anaerobic respiration. [6]

- 1 conversion of pyruvate to acetyl CoA (Link Reaction), the Krebs Cycle and oxidative phosphorylation are blocked.
- 2 because electronegative oxygen, which is the final electron acceptor not available to pull electrons down the electron transport chain
- 3 glycolysis oxidises glucose to 2 molecules of pyruvate.
- 4 oxidising agent of glycolysis is NAD⁺, not oxygen.
- 5 glycolysis generates 2 ATP by substrate-level phosphorylation whether oxygen is present or not.
- 6 for glycolysis to continue under anaerobic conditions, a sufficient supply of NAD⁺ must be available for accepting electrons during the oxidation step of glycolysis.
- 7 regenerate NAD⁺ by transferring electrons from NADH to pyruvate during fermentation process